



Geosynchronous transfer orbits as a market for impulse delivered by lunar sourced propellant

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ABSTRACT

Every year around 100 metric tons of commercially addressable geostationary satellites are launched into geosynchronous transfer orbits (GTOs). The orbit raising from GTO to geostationary equatorial orbit (GEO) is an existing market for impulse that could be addressed by a service using propellant sourced from the Moon. We reconstructed the economic model in the recently published *Commercial lunar propellant architecture: a collaborative study of lunar propellant production* (Kornuta et al., 2019) and used it to investigate the financial viability of providing propulsion as a service in GTOs. We used NASA's General Mission Analysis Tool to design a GTO service delivery flight plan, which we extend to cover sub and super synchronous delivery. We used economic, astrodynamics, and engineering models to derive the sensitivity of the internal rate of return to GTO launch costs, sale prices, tug inert mass fractions, and delivery orbit. While falling launch prices erode the impulse service returns, they may create other markets for propellant, we explore some higher return products these markets would make available.

1. Introduction

The potential for lunar resource utilization has been mooted since before humanity set foot on the moon. Large scale possibilities were explored in the context of space settlements and space solar power satellites during the seventies, for example Bock (1979). Blair et al. (2002) presented detailed economic modeling of polar lunar water derived propellant to service the market for lifting satellites from LEO to GEO. In 2004 the U.S. President's "Vision for Space Exploration" called for National Aeronautics and Space Administration (NASA) to "develop ways to utilize the Moon's resources to support manned exploration". Sanders and Larson (2012) note that "Incorporation of in situ resource utilization (ISRU) and the production of mission-critical consumables for propulsion, power, and life support into mission architectures can greatly reduce the mass, cost, and risk of missions" and detail ongoing work to develop technologies and hardware to realize this vision. Oeftering (2011) and Gill et al. (2012) detail lunar derived propellant based cis-lunar transport architectures supporting science, exploration, and commerce; including lifting satellites from LEO to GTO and from LEO to GEO, respectively. Labriet and Poulet (2013) identified commercial communications satellites as the most profitable existing space industry and noted the potential for using lunar derived propellant for LEO to GEO impulse. More recent studies explore the beneficial impact of lunar ISRU for Mars exploration (Ishimatsu et al., 2016), and the potential demand

generated (McVay et al., 2016).

Recent increases in the probability of water ice in polar lunar craters (Nozette et al., 2001; Feldman et al., 2000; Mitrofanov et al., 2012) (perhaps most notably the spectroscopic analysis of the LCROSS impact ejecta plume finding $5.6 \pm 2.9\%$ H₂O by mass Colaprete et al. (2010)) prompted the publication of a collaborative paper detailing a commercial approach to satisfying potential demands for propellant in cis-lunar space (Kornuta et al., 2019). Both Blair et al. (2002) and Kornuta et al. (2019) provide economic analysis as a measure of utility, a better economic case indicates that in a noncommercial context less subsidy/funding is required. Both papers use internal rate of return (IRR) as a figure of merit, IRR allows the direct comparison of proposals with different scale and timing.

Water can provide a complete LH₂/LOX propellant, but when burned in the non-stoichiometric ratios usually employed by rocket engines there are significant amounts of excess oxidizer byproduct, the water:LH₂/LOX mass ratio is approximately 3:2. Where water is absent there are proposals to use oxygen/silane (Rosenberg, 1985), or even oxygen/metal (Hepp et al., 1994) propellants. The literature also explores the potential for oxidizer alone, with Hepp et al. (1994) noting predictions of 40%–60% reductions in initial mass in LEO from lunar oxidizer used with terrestrial fuel.

In this paper we draw on the current lunar propellant work of Kornuta et al. (2019) and follow Blair et al. (2002), Oeftering (2011) and Labriet and Poulet (2013) in analyzing an impulse service to GEO, but we

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Nomenclature

ACES	Advanced Cryonic Evolved Stage
EML1	Earth Moon Lagrange Point 1
EBITDA	Earnings before interest, tax, depreciation and amortization
FAA	Federal Aviation Administration
GEO	Geosynchronous Equatorial Orbit, Geostationary Orbit
GMAT	General Mission Analysis Tool (NASA)
GSO	Geosynchronous Orbit
GTO	Geostationary Transfer Orbit
IMF	Inert Mass Fraction
IRR	Internal Rate of Return
I_{sp}	Specific Impulse
ISRU	In Situ Resource Utilization
LEO	Low Earth Orbit
LH2/LOX	Liquid Hydrogen/Liquid Oxygen propellant
NASA	National Aeronautics and Space Administration
SSTO	Supersynchronous Transfer Orbit
TEI	Trans Earth Injection
TESS	Transiting Exoplanet Survey Satellite

propose serving the demand in GTOs rather than LEO. GTOs are cheaper to reach from the Moon, more expensive to reach from Earth, and are where current GEO satellite launch services deposit their payloads. These significant advantages are potentially challenged by orbital dynamics and satellite launch constraints. We design a flight plan that imposes operational constraints on satellite launches that are like current launch practices and derive a delta-v budget for the impulse delivery flights using NASA's General Mission Analysis Tool (NASA, 2019). We also explore the financial implications of using mined propellant to transport excess oxidizer byproduct for sale. We combine financial, astrodynamic, and engineering models to derive the sensitivity of returns to launch prices, sale prices, delivery orbits, and vehicle inert mass fractions. Given the uncertainty regarding the disposition of lunar water, mining techniques, and of market penetration one cannot argue that our estimates of returns/utility are accurate. However, we show that GTO delivery is achievable, has higher utility than LEO delivery, and we provide a sensitivity analysis that allows one to explore cost and market scenarios.

We conclude that the business case for the delivery of GEO satellite impulse to GTOs is much stronger than for LEO, and that using the mine costings of Kornuta et al., (2019) it can generate significant savings for GEO satellite launches. There is potential for similar benefits to science and exploration missions; however interplanetary departure orbits are more constrained than a GEO satellite launch, so further work is needed to characterize that potential. We also show that in a propellant market supplied by a water to LH2/LOX mine one can achieve higher returns by using the produced propellant to transport the excess oxidizer for sale.

Table 1

Important prices, costs, and consumption for Earth/Moon propellant.

	LEO	GTO	GEO	Moon	
Launch from Earth cost	4000	8000	16,000	35,000	\$/kg
Sale price \$/kg	3000	6000	12,000	27,000	\$/kg @ 25% discount
From Moon gear ratio	6	3	3	1	kg @ Moon/kg delivered
From Moon	5	2	2	0	kg burnt/kg delivered
Current Consumption		60			Tons of Bi-Propellant (on satellite at separation)
Potential Consumption	210	45			Tons of LH2/LOX

The first three rows are sourced from Table 11 and Figure 70 of Kornuta et al. (2019). The derivation of the GTO Consumption figures from FAA data are detailed later. The LH2/LOX LEO Potential Consumption is from Table 12 of Kornuta et al. (2019). It should be noted that these are abstractions based on assumptions about propellant used, vehicle engineering, and detailed trajectories. We initially use these figures to retain comparability with Kornuta et al. (2019), and later derive ratios for the Moon to GTO case with specific vehicles in Section 6.

2. Lunar water as propellant; a recent proposal

There is good evidence that permanently shadowed regions of the lunar poles contain recoverable water. This water can be liberated, processed to propellant, and distributed to customers via a logistics network of tugs and orbital depots. The recently published *Commercial lunar propellant architecture: a collaborative study of lunar propellant production* Kornuta et al. (2019) details an integrated and collaborative approach to lunar sourced propellant. The paper provides financial modeling for seven scenarios ranging from "LEO to GEO", through to large manned exploration programs. We use the financial model and assumptions from this paper to examine a GEO satellite orbit raising service.

A key financial and engineering driver is that moving mass up or down the Earth or Moon gravity wells consumes propellant. A lunar water mine must produce both payload and transportation propellant. The ratio of the initial propellant mass at launch from the Moon to the propellant delivered at a location is referred to in Kornuta et al., (2019) as the "gear ratio". A higher gear ratio means a larger scale mine with a higher investment in mining operations.

One straightforward way to derive the gear ratio for delivering payload mass using a single stage with given inert mass is to begin by rearranging the rocket equation to compute the propellant required to impart a given delta-v to a final mass:

$$m_p = m_f \left(e^{\frac{\Delta v}{v_e}} - 1 \right) \quad (1)$$

Where m_p is the propellant mass, m_f is the final mass, here Δv is 3400 m/s delta-v from Moon to GTO and the same for the return (see Table 4) and v_e is exhaust velocity of 4413 m/s (450 I_{sp}). The tanker will need to make a round trip with a total delta-v of 6800 m/s but leaving payload mass at GTO. First, set m_f to be the mass of the tanker and compute the return propellant required for the GTO to Moon leg. Next, set m_f equal to the mass of the tanker, return propellant, and payload to compute the propellant required at lift off. Finally, one computes the gear ratio from total propellant and payload mass, and thus the inert mass fraction. Table 2 shows a result where we numerically optimized an inert mass to achieve 3:1 gear ratio, it requires an IMF of 7.1% or lower.

From the sale price and gear ratio in Table 1 a supplier wants to sell propellant on the Moon itself; zero kg/kg spent in transport and the highest price. However, there is no current market on the Moon. GTO looks interesting because, relative to LEO, one spends less than half the transportation propellant but doubles the income per kg delivered. This means less than half the mine costs (or double the saleable kilograms).

3. GTOs: an existing market for lunar propellant through impulse delivery

Every year around 100,000 kg of US company addressable satellites are placed in GSOs. (derived from FAA (2018)). Orbit raising from LEO has been proposed as a propellant market (Blair et al., 2002; Labriet and

Table 2

Example derivation of Gear Ratio for GTO delivery of propellant from the lunar surface.

Leg	m_p	m_f	m_{payload}
GTO-Lunar Surface	0.2656	0.2288	
Lunar Surface-GTO	1.7346	1.4944	1.0000
At Launch	2.0002	3.2290	
Tanker IMF		0.0709	

Here we set $m_{\text{payload}} = 1$ and numerically solved for GTO-Lunar Surface m_f such that At Launch m_p (the sum of the leg m_p s) was 2. That delivers a 2:1 burn:deliver ratio and allows computation of the tanker IMF.

Table 3

Comparison of LEO and GTO markets.

	Propellant market, tons	Annual revenue, millions	Water required, tons
LEO	210	\$630	1880
GTO	45	\$360	205

GTO: 57% of the revenue from 11% of the water. The LEO data is from Kornuta et al. (2019) LEO propellant case Table 12. GTO data is using 2:1 burn ratio, and 3:2 water:LH2LOX ratio. All the revenues are calculated using Kornuta et al. (2019) Figure 70.

Poulet, 2013; Oeftering, 2011) and possibly Kornuta et al. (2019); however, for a prospective lunar supplier:

1. The sale price is low
2. The gear ratio is high, product is expensive in propellant to deliver
3. Current launch procedure is direct to GTO, with the satellites then raising their own orbit
4. There are no vehicles that are “refuellable in orbit”; either satellites or tugs

Since there are no on orbit refuellable vehicles, strictly speaking there can be no current demand or market for propellant. This also leads to a chicken and egg situation where prospective producers and consumers wait on each other to act.

Although current GEO satellites cannot accept propellant, they could accept impulse in GTO. Currently, this impulse is provided by onboard “apogee motors” (they fire at apogee to raise the perigee), but the impulse could be delivered by a tug. Since any lunar water to propellant mining operation will have tankers for moving water and/or propellant they would also have the capability to supply impulse using those tankers. The potential for providing propellant or impulse to GTO was not explored in Kornuta et al. (2019), however, the paper provided enough detail for their financial model to be reconstructed and applied to this new scenario.

GSO satellites are lifted on a variety of launchers and propellants directly into a GTO. Some launch services, like Proton, use an upper stage to do partial orbit raising and inclination correction; this is to enable them to be competitive with lower inclination launch sites. The industry rule of thumb is that the launcher leaves about 1500 m/s for the satellite

Table 4

Delta-v budget for GTO impulse delivery from lunar poles.

Flight plan step	Note	m/s	Source
1 & 10 (each)	Polar LLO & Surface	1870	Thronson (2008); Jones et al. (2011) & checked by vis viva & Hohmann transfer from LLO to surface
2	TEI	842	GMAT – allowed the optimizer to find the transfer to a 185 km perigee with 28-degree inclination.
3 & 5 (total)	Nominal GTO,	<678 <	Vis viva – lowering lunar distance apogee to GEO altitude.
6	supersynchronous is less	<2098 <	Vis viva – doing orbit raising and inclination change to GEO altitude.
7	TLI	1067	GMAT – allowed the optimizer to find a TLI and capture burn to match the first stage inclination.
8	Lunar capture	780	
9	Polar inclination change	57	Residual inclination mismatch from GMAT lunar capture, computed via circular orbit inclination change.

Steps 3, 5, & 6 will vary depending on the apoapsis of the delivery orbit. Step 4 is the launch of the GEO satellite from Earth, and so is not part of the delta-v budget for impulse delivery.

to complete (Belo, 2019); with less mass efficient storable propellants, or less time-efficient electric engines.

Assuming an annual 156,000 kg of separation mass (based on a conservative interpretation of the FAA data (FAA, 2018)) and 1500 m/s of delta-v imparted with 320 I_{sp} storable propellant we can use the rocket equation to derive the annual commercially replaceable storable propellant to be around 60,000 kg. At the prices in Table 1 that is a \$360 million market. One can supply the same impulse with about 45 tons of 450 I_{sp} LH2/LOX. Given the gear ratio in Table 1 that means a mine water output of approximately 205 t/year.

The comparison with the LEO case from Kornuta et al. (2019) shows that the GTO market is worth examining in more detail; a GTO market delivers over half the revenue with about a tenth of the water production (See Table 3). All else being equal the thermal tent water mining and propellant production methods of Kornuta et al. (2019a) seem likely to scale linearly. Firstly, the energy requirements are dominated by the mass of water sublimated and electrolyzed. Energy is provided by solar arrays or heliostats, which scale linearly with power output. Secondly, the individual transformation and transportation processors are envisaged as each handling only a fraction of the total production volume, so we can downscale by using fewer elements. If smaller production volumes require more mass than is implied by a linear relationship it would erode our case; but given the current state of knowledge, it seems premature to try to derive more exact scaling laws. As we detail later, we have identified and handled some nonlinearities; we assume the same operational expenses as Kornuta et al. (2019) proposes for the large scale mine, and we add explicit hardware costs for our tugs.

The revenue and costs from Table 3 make a compelling case for the GTO market, we next detail how the GTO orbit raising service would function, which shows how the proposal of Kornuta et al. (2019) can be downscaled and repurposed. We then do an astrodynamics validation of the burn to deliver ratio, use the financial model of Kornuta et al. (2019) to derive financial performance, and produce sensitivity analysis.

4. GTO to GSO orbit raising as an impulse service

We take the Kornuta et al. (2019) proposal and modify it to provide only impulse to satisfy the GTO market. We begin here by sketching the concept of operations and its advantages. In Section 6 we do an engineering and astrodynamics analysis for feasibility, and in Section 7 we use that in an economic analysis.

1. Impulse is supplied as follows:

- a. A propellant tug launches from the polar lunar mine/depot.
- b. The tug places itself in an agreed supersynchronous GTO and makes phasing maneuvers to prepare for a timely rendezvous with the payload.
- c. A GSO bound payload launches to the agreed GTO.
- d. The tug rendezvous and grabs the payload (either via a multi docker like Northrop Grumman’s Mission Extension Vehicle (NorthropGrumman, 2019), or graspers as in Kornuta et al. (2019) robotics section).

- e. The tug raises the payload perigee and deposits it in its destination GSO.
- f. The tug returns to the lunar pole mine/depot.
2. Downscale the mining operation to produce only 205 t/year of water.

Packaging the propellant as a service has several advantages:

1. It turns the abstract potential market for propellant into an existing impulse market. GEO satellites currently pay for impulse through onboard engineering and would pay for an orbit raising service.
2. It does not require major changes to GEO satellite launch practices or engineering.
3. It requires no standards agreement, engineering, or research around microgravity refueling, nor orbital depots.
4. It provides a low risk incremental adoption path. The GEO satellite launch business is conservative, for example the adoption of electric drives has been quite slow and incremental:
 - a. On occasion a GEO satellite launch will partially fail, for example the US Navy MUOS in 2016 ([Spaceflight101, 2016](#)), or ChinaSat 9A in 2017 ([Gunter](#)). This can leave the satellite irrecoverable or having to sacrifice a large portion of its lifetime station keeping propellant to achieve the destination orbit, both previous examples discarded 10 years of their 15-year lifespans to achieve the destination orbit. The five satellite MUOS program cost \$7.4 billion ([News, 2015](#)), so although such salvage operations would be unpredictable income, they could be a lucrative way to prove the service and provide ongoing windfalls.
 - b. A partial orbit raising of an unaltered satellite has utility by extending station keeping lifetime, in the case of electric engines the satellite can also be operational earlier. This partial raising can provide utility at any point in the satellites normal apogee raising.
 - c. As reliability is proven satellites can begin to be under fueled to reduce launch mass.
 - d. When the service is accepted as fully reliable satellite apogee raising capability can be eliminated altogether. This can translate to larger satellites or more ride sharing.
5. One can also use the impulse for life extension by taking responsibility for moving satellites to graveyard orbits, freeing more onboard propellant for station keeping.

5. LOX as a product

Producing LH2/LOX rocket propellant from water electrolysis generates a significant quantity of excess oxygen. In most studies this is discarded, with indications that future life-support or other uses may eventuate. In contrast many studies ([Rosenberg, 1985](#); [Hepp et al., 1994](#); [Schreiner, 2015](#)) deal with the production of just oxidizer where there is no prospect of hydrogen extraction, for example by Molten Regolith Electrolysis ([Schreiner, 2015](#)). We explored the financial implications if a water to propellant mine treated the excess oxidizer as its primary product.

Water is composed of hydrogen and oxygen in a 1:8 mass ratio, LH2/LOX rockets burn a mass ratio of between 1:5 to 1:3.5. This means if one produces propellant by electrolyzing water, then from a third to a half of the mine output by mass is byproduct and discarded. This is equivalent to oversizing the mine, the largest cost to the business, by a factor of 1.5 or 2. One could design rocket motors to burn closer to a stoichiometric ratio which would increase mine output utilization at the cost of lower I_{sp} . However, we did not pursue this avenue since it is a non-trivial engineering project. However, LOX is used by many propellants and customers; with hydrogen (ULA), methane (SpaceX), and natural gas (Blue Origin) for example. By mass CH₄/LOX is 78% LOX and LH2/LOX is 78%–83% LOX, so propellant customers would certainly derive a benefit from purchasing LOX only. The 2009 NASA Mars Design Reference Architecture ([Drake et al., 2010](#)) broke with earlier versions by switching the Mars ascent vehicle from ISRU methane and Earth sourced oxidizer to

ISRU oxidizer and Earth sourced methane.

Oxidizer could be sold wherever propellant could be sold, for example at a cis-lunar aggregation point (as in [Kornuta et al. \(2019\)](#)) or in highly elliptical GTO like orbits as part of an interplanetary distributed launch/perigee kick mission plan.

Propellant produced by the water mine operation would be used for transporting LOX to the point of sale, depending on fine details of burn ratios there might be some residual LH2 available. Since the main determinant of the sale price of a product in orbit is the competitive cost of raising it from earth one can expect to sell LOX for the same price one would sell propellant. By treating LOX as a product, one immediately increases mine revenue by 50% in EML1.

6. Verifying the propellant “burn to deliver” ratio

The preceding economic/utility arguments are compelling, but it is not immediately obvious that there are flight plans that achieve the “burn two, deliver one to GTO” gear ratio of [Kornuta et al. \(2019\)](#). Also, in the case of impulse delivery the tug needs to raise itself to GEO before returning to the Moon, adding extra costs above those of propellant delivery to GTO.

Challenges include; the inclination of the Moon’s orbit relative to the satellite launch GTO, and the angle of the line of apsides of the launch GTO and the tug’s Earth injection orbit. For a given launch site there is an optimal launch inclination. For example, Kennedy Space Center launches into orbits with a 28-degree inclination. There is an infinity of orientations for a 28-degree orbit depending on the launch time of day. Inclination change is very expensive, one reduces the cost by doing the maneuver when orbital velocity is low and combining it with some other maneuver, so GTOs generally place their apoapsis on the equatorial plane. Changing the line of apsides is also expensive; current launches target for the line of apsides to provide a good eclipse environment, to ensure batteries can maintain charge. The apoapsis can range from subsynchronous to supersynchronous (SSTO), below or above geostationary altitude.

We explored a few options and found the following flight plan that looks achievable without placing prohibitive constraints on the GEO satellite launch:

1. As the Moon is approaching the Earth equatorial plane a two-stage tug launches to a lunar polar departure orbit. Since the Moon’s rotation is so slow the orientation of this orbit can be freely chosen to minimize the trans Earth injection maneuver.
2. The second stage separates and performs a trans Earth injection:
 1. The polar orbit orientation and maneuver are chosen to place post injection perigee, and thus the line of apsides, on the line of nodes of the Earth equatorial plane and the lunar orbit plane.
 2. The inclination is matched to the satellite launch site inclination.

At perigee the tug does a small burn to lower its apogee, placing itself in a highly elliptical, supersynchronous transfer orbit like, phasing orbit. To minimize propellant expenditure the tug requires all maneuvers prior to rendezvous to monotonically lower its apogee, so the tug must phase with higher apogee than the client GEO satellite. As GEO satellite launch time approaches the tug performs phasing burns so it is poised to rendezvous in a timely manner.

4. The GEO satellite launches to a GTO that matches the line of apsides of the tug. This is operationally like choosing the line of apsides for eclipse properties.
5. We modelled a perigee rendezvous, followed by apogee impulse delivery. The tug can more cheaply reach supersynchronous apogees.
6. The tug performs a conventional orbit raising and inclination correction, phasing to the correct GEO slot. Finally, the tug releases the satellite.

7. The tug now performs small phasing burns and waits for an appropriate time to do a trans lunar injection from approximately GEO, to encounter the Moon at the opposite node it departed from.
8. The tug captures to a polar orbit which is within a few degrees of its original departure orbit.
9. The two stages match planes and rendezvous, restacking the stages by berthing.
10. The tug lands using residual first stage propellant, ready for reuse.

There may be lower delta-v segments one could employ; for example, GTO to the Moon using four body effects have been explored and might be usefully reversible (Genova et al., 2015). Also using a highly elliptical lunar polar orbit might reduce total propellant usage by reducing the TEI burn of the second stage. However, the flight plan above is straightforward to work with and can be easily tweaked to cover various sub and supersynchronous transit orbits by using the vis viva equation on steps 5 and 6.

We briefly review the constraints this flight plan imposes:

- i. There are two opportunities a month for lunar tug launches, per the FAA compendium (FAA, 2018) there are around 20 orbit raises per year, so this constraint does not seem too onerous.
- ii. The constraint placed on the GEO launch is also not as onerous as it may seem. For a launch with a coast in LEO there is one launch window a day. The launch time places the LEO in the correct plane and then there are many opportunities a day to push apogee out in the correct direction.
- iii. The return lunar orbit matching the departure lunar orbit is important for a two-stage tug with the stages reuniting in lunar orbit. As it turns out the departure and arrival lunar orbits naturally align to within a few degrees.

6.1. Further work

- i. A staging scheme where the upper stage always remains in lunar polar orbit is also possible. It could have a somewhat lower propellant requirement (the IMF reduction for a pure micro gee tug vs the tankage increase on the lander); but with an additional rendezvous on propellant launch, and micro gravity propellant transfer.
- ii. The lunar polar departure orbit cannot really be “freely chosen” but rather “cheaply chosen”. The departure orbit can probably also be compensated for by the timing and size of the TEI. There is a trade space here to explore.

Many of the flight plan segments above can be analyzed simply using published numbers or computed via the vis viva equation, but the trans Earth injection and lunar capture are more complicated. We used NASA’s General Mission Analysis Tool (GMAT) to do a detailed design of an instance of the TEI placing the post TEI perigee at 185 km and inclination 28°, and similarly modeling the return to the Moon from GEO. GMAT does high fidelity orbit propagation with detailed gravitational anomaly models. It is operationally certified; its outputs can be directly used by operating spacecraft - it was the primary mission design tool for Transiting Exoplanet Survey Satellite (TESS) (NASA, 2019).

While a 28-degree inclination is between Kourou’s 5° and Baikonur’s 49°, the above flight plan is a conservative estimate for the delta v given industry expectations of a 1500 m/s orbit raising (Belo, 2019).

We use the above delta-v budget, the rocket equation, and tug stage inert mass fractions to derive the propellant available for providing satellite orbit raising - which gives us the burn to deliver ratio.

Inert mass fractions (IMF) of 0.26 or more can easily be found for lunar landers like Altair (Jones et al., 2019) or “concept 5 crew taxi” at 0.30 (Kennedy, 2008), but we feel these are not good proxies for a propellant tug. Many lunar landers must survive Earth launch fully laden, or

are human rated and pressurized, or use storable propellants, or need to land and offload heavy cargos. Something like a Centaur/ACES seems a more reasonable proxy (Barr, 2015), it is used as the basis for the proposed XEUS lunar lander (Kutter et al., 2009). One can achieve a tank mass 7% of the propellant mass for 30 metric ton LH2/LOX vehicles by using ALi 2195 alloy (Pietrobon, 2009). Composite tanks might deliver a further 30% reduction of tank mass (Knapschaefer, 2016). Optimizing the design for cis-lunar operation might further reduce mass; due to reduced launch loading and not needing to deal with the launch thermal, atmospheric, and aerodynamic environment.

Boiloff of cryogenic propellant is a significant factor and there will be a trade space around boiloff management mass, mission duration, and acceptable boil off rate - all contributing to the “all-up gear ratio”. One can use the IMF and burn:deliver ratios below to make an informal allowance for boiloff.

We chose IMFs 12% for stage 1 and 10% for stage 2 as baselines, then produced two more variants by increasing each stage IMF by a third. Finally, we include an ultra-light second stage as might be technically feasible and/or useful for large scale oxidizer delivery to highly elliptical orbits to support distributed launch for interplanetary spacecraft. Our baseline IMFs were chosen optimistically but without any investigation of the ultimate potential of optimization for cis-lunar operation.

From the above we see that one should be able to burn two to three units of propellant for every unit delivered to supersynchronous or GTO orbits respectively. One can also see that the burn ratio is much more sensitive to stage 2 IMF (5500 m/s and added satellite mass for a portion of that) than to stage 1 IMF (3800 m/s); we explore the financial implications below.

7. A financial model for impulse in GTO

We reproduced the finance model of Figure 71 in Kornuta et al. (2019) and verified our reproduction against their scenarios, this allows our GTO impulse case to be directly compared with the cases modelled in Kornuta et al. (2019). We minimize any perturbation of the Kornuta et al. (2019) assumptions in order to facilitate comparison. All the mining efficiency parameters, hardware cost, and transportation costs could be challenged or altered but we are aiming to isolate the financial effect of delivering impulse to a GTO. We begin by using a 2:1 burn ratio to illustrate our process. We have shown, in Fig. 2, that tug engineering and orbital assumptions determine the burn ratio, so we later model a range of ratios (see Fig. 2). For rescaling the mine, the key parameters are:

- Mine equipment kg/(mined water kg/year) = 0.0121951 (30,000 kg mine producing 2,460,000 kg of water per annum)
- Annual parts replacement fraction = 0.026666 (30,000 kg mine requiring 800 kg per annum).

An annual GTO demand for 45,000 kg LH2/LOX to replace 60,000 kg of storable propellants, delivered at a 2:1 burn to deliver ratio translates to a requirement for 205 t of water on the lunar surface. The financial model in Kornuta et al. (2019) determines:

- 2497 kg of mining equipment at \$100,000/kg + \$35,000/kg transport to a lunar pole.
- demand for 60,000 kg of replaced propellant in GTO at \$6000/kg

The resulting mine size is too small to include 3 sets of tug masses so we added tugs to the model; each stage priced like a Centaur (though they would be smaller) and transported to the Moon at the baseline \$35,000/kg, this effectively doubles the cost base of Kornuta et al. (2019). Replacement and repairs are a difficult area to model and the approach we have taken is to follow Kornuta et al. (2019), more comprehensive modeling will only be required if the returns of GTO impulse are sufficiently compelling.

Table 5 shows a baseline to baseline + tugs comparison of the LEO

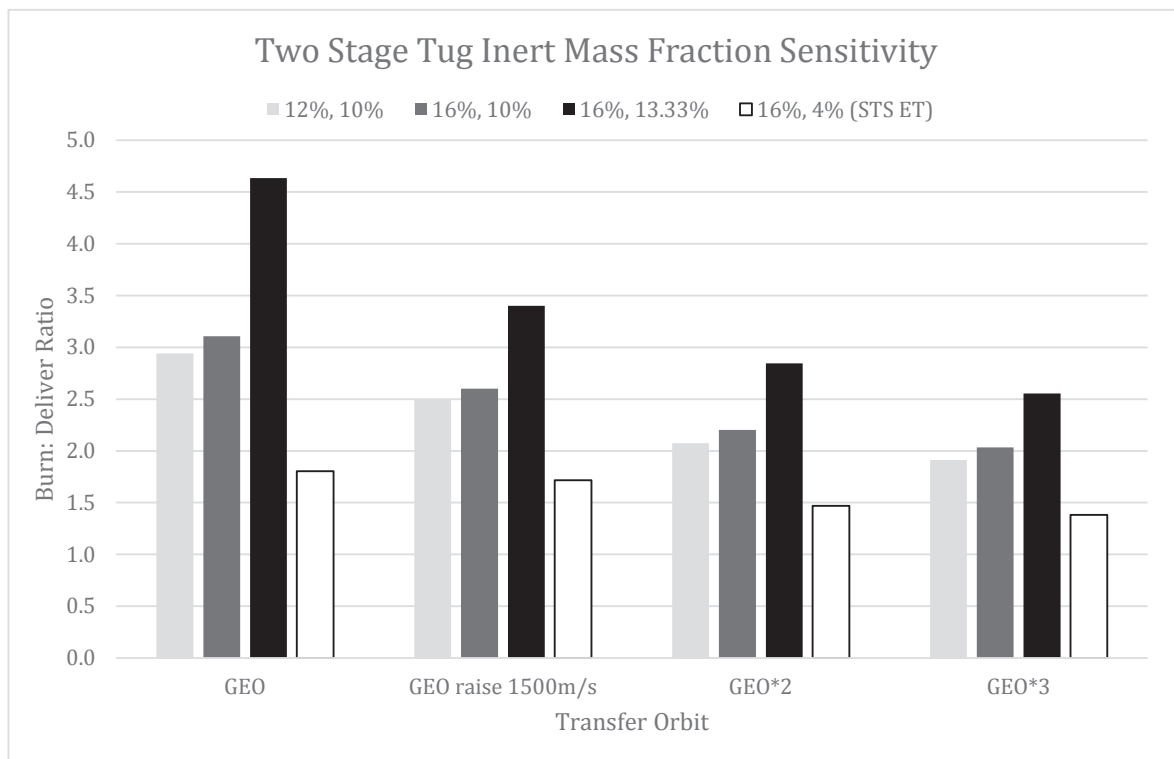


Fig. 1. The burn to deliver ratio achieved by various inert mass fractions of a two-stage propellant tug. The four clusters represent variations on our baseline orbit raising from a 28-degree GTO to GEO with different apogee altitudes and residual delta-v to raise; baseline, only 1500 m/s of orbit raising (per operator rule of thumb), a supersynchronous apoapsis (2*GEO), and a highly supersynchronous apoapsis (3*GEO). All these transfer orbits have been employed by past orbit raisings.

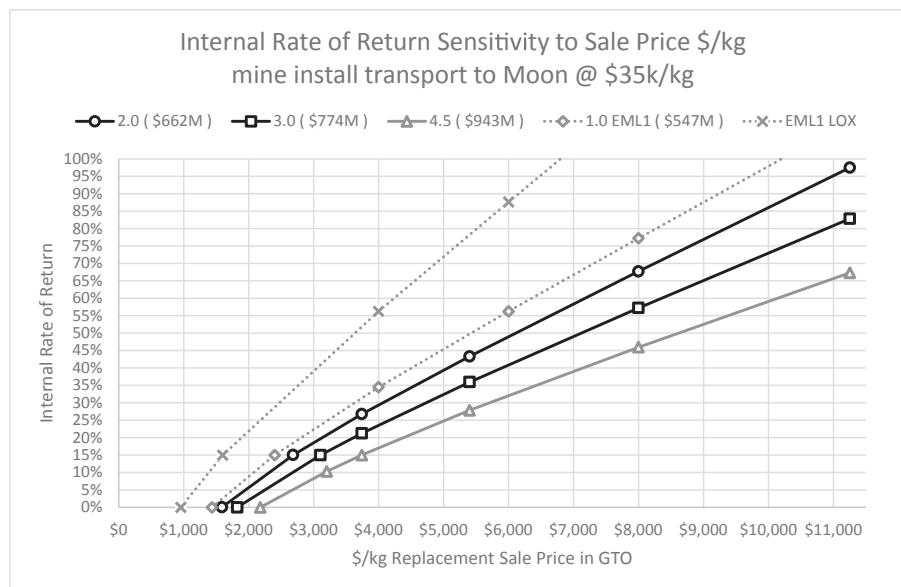


Fig. 2. IRR against sale price for various burn ratios; solid series are for GTO delivery of impulse; dotted series are for EML1 delivery of propellant. The initial investment to support a given series is noted in the legend. One might achieve a given burn ratio by different combinations of tug engineering and delivery orbit apogee. Fig. 1 details some examples for GTO impulse from which we selected the plotted burn ratios. The EML1 ratios are taken from Kornuta et al. (2019).

and GTO markets, using all the ratios and discounts of Kornuta et al. (2019). The internal rate of return (IRR) for GTO is much higher, higher than all but one of the seven scenarios analyzed in Kornuta et al. (2019).

By focusing on just the existing demand for impulse in GTO one can achieve much better financial performance than the hypothetical LEO market.

We modelled burn ratios of 2, 3, & 4.5 to produce a sensitivity

analysis of the internal rate of return (IRR) to the sale price for replacing a kg of storable propellant in the delivery orbit. We also modelled propellant and LOX sale at EML1, here the mine is sized to produce 90 t of LH2/LOX on the lunar surface to address a hypothetical EML1 market that can consume the entire mine output. Often (as in Kornuta et al. (2019)) launch to LEO \$/kg is used as a sensitivity axis however the impulse service is competing against launch to GTO. Since GTOs are

Table 5
Comparison of baseline LEO vs GTO propellant sale scenarios.

	(a) 210,000 kg/y @ LEO (from Scenario 3 in Kornuta et al. (2019))	(b) 45,000 kg/y @ GTO (replacing 60,000 kg/y of storable propellant at \$6000/kg)	b/a as percent or \$B change
Initial Investment	\$4.050	\$0.662	16%
Revenue	\$0.630	\$0.360	57%
EBITDA	\$0.502	\$0.331	66%
IRR	4%	49%	
NPV @ 10%	-\$0.966	\$1.371	+\$2.337

\$ amounts in billions.

achieved by a variety of propellants ranging from LH2/LOX (ULA) through to storable propellant (Proton) a launch to LEO price does not translate directly to a GTO price.

The series for the EML1 market are only indicative; we did not verify the burn ratios or tune tug sizes. These two series represent the sale of propellant, not an impulse service. The reduction in the cost of transport to the market allows a smaller mine size, which produces most of the increase in IRR. The EML1 LOX series takes the EML1 mine and uses all the hydrogen to transport LOX for sale in EML1. This change in strategy increases the effective mine output, and thus revenue, by 50%.

One can use Figs. 1 and 2 together to explore market and risk scenarios. From Fig. 1, choose an inert mass fraction and delivery orbit to imply a burn ratio. For example, the medium tugs to a standard GTO and 1500 m/s required for orbit raising implies 2.5 units burnt for every unit delivered. From Fig. 2, one can choose a competitor GTO launch cost, apply a discount to arrive at an achievable sale price, and lookup the return. One can then adjust the IRR for risk and compare to a hurdle rate. For example, SpaceX is currently charging just over \$11,000/kg to GTO (SpaceX, 2019). Discounting to an \$8000/kg sale price at a 2.5:1 ratio would generate an IRR of just over 60%. Applying risk and funding discounts totaling 30% still leaves a significant 30% residual return.

In a non-commercial context, one can choose a perceived risk level and lookup the replacement price. For example, with 10% risk discounting one should be able to replace storable propellant for \$2700/kg in a GTO with baseline tugs which scales to \$3560/kg to replace LH2/LOX. The likely use case for this scenario is a large science or exploration mission, which would have more constrained target trajectories and larger delta-v demands than GEO satellite orbit raising. Our current analysis is not focused on these more challenging cases, but it seems there is some potential here.

There is uncertainty and risk around mining techniques and technologies, and thus around the cost of implementing a mining operation. Fig. 3 selects two interesting return rates and plots them in \$/kg replacement sale price and \$/kg hardware cost multiplier; 15% represents a hurdle rate, and 30% a risky hurdle rate (a 15% risk discount rate compounded with the hurdle rate). Note that we scaled all the hardware costs, but about half the baseline costs are for well-understood tugs so the estimates are conservative. If one thought an \$8000/kg replacement sale price was achievable for a 2.5 burn to deliver ratio, then hardware costs could be \$250,000/kg and return 30% IRR.

8. Conclusion

We have shown that there is an existing market for off Earth propellant when embodied as a GEO satellite orbit raising service, an impulse in GTO service. Recasting a propellant supply enterprise into an impulse service avoids the chicken and egg dilemma, where producer and consumer must each rely on the future behavior of the other. An orbit raising service can simply compete directly with existing service providers for existing customers. Using data from (FAA, 2018), the financial model from Kornuta et al. (2019), and achievable flight plans, we have shown that an impulse service can generate high returns/utility in the

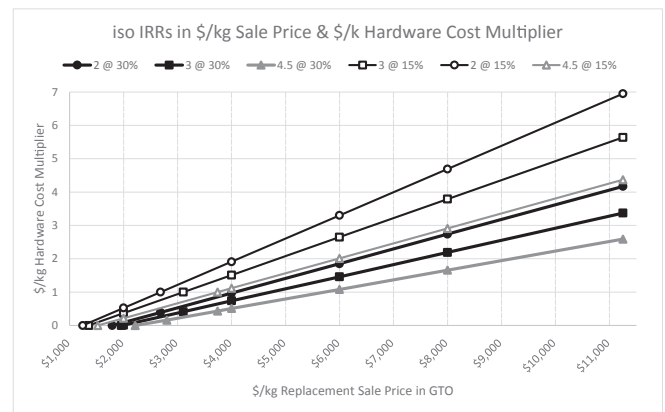


Fig. 3. Lines of iso IRR for replacing storable propellant in GTO plotted in \$/kg sale price and \$/kg Hardware Cost Multiplier. The hardware cost multiplier is applied to the \$100,000/kg from Kornuta et al., (2019) that we used to derive our Fig. 2.

face of variations in sale price, destination orbits, and engineering assumptions.

We have also shown that when a market for propellant exists, then one can make significantly higher returns by using the propellant for transporting and selling oxidizer, a byproduct that is usually discarded. The business case of an impulse service is threatened by lower Earth launch costs, which reduce revenues. However, if lower launch costs help open a market for propellant, then the impulse mine can switch to the sale of oxidizer and make better returns than its original business case.

The business cases of impulse service and propellant supply are more easily closed in GTOs, and GTO like orbits, than in LEOs. These business cases are a reachable steppingstone toward wider use of off-Earth propellant.

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Nicholas James Bennett: Conceptualization, Methodology, Software, Visualization, Writing - original draft. **Andrew G. Dempster:** Supervision, Writing - review & editing.

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